

MAKE UP EXAMINATIONS – JUNE 2023

Program	: B.E :- Common to CSE (CY) / CSE (AI & ML)	Semester	: III
	: / (AI & ML) / (AI & DS)		
Course Name	: Discrete Mathematical Structures	Max. Marks	: 100
Course Code	: CI35 / CY35 / AI34 / AD34	Duration	: 3 Hrs

Instructions to the Candidates:

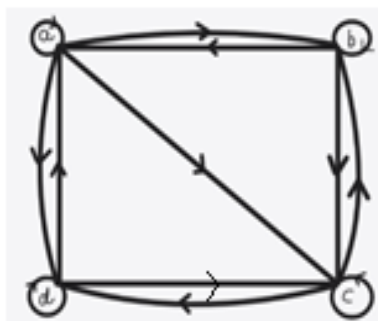
- Answer one full question from each unit.

UNIT - I

- Give a direct proof that if m and n are both perfect squares, then nm is also a perfect square. CO1 (06)
 - Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent without using truth table. CO1 (07)
 - Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology without using truth table.
 - Use mathematical induction to prove that $7^{n+2} + 8^{2n+1}$ is divisible by 57 for every nonnegative integer n . CO1 (07)
- Show that $\neg \forall x(P(x) \rightarrow Q(x))$ and $\exists x(P(x) \wedge \neg Q(x))$ are logically equivalent. CO1 (06)
 - Determine whether each of the compound propositions:
 $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ and $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is satisfiable. CO1 (07)
 - Show that at least four of any 22 days must fall on the same day of the week by the method of contradiction. CO1 (07)

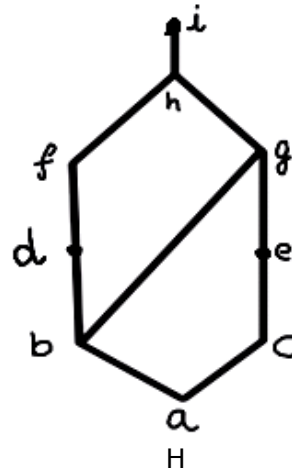
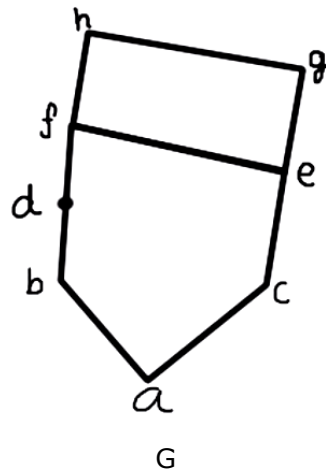
UNIT - II

- Determine whether the relation with the directed graph shown below is an equivalence relation. Justify. Also write the matrix of the relation. CO2 (06)

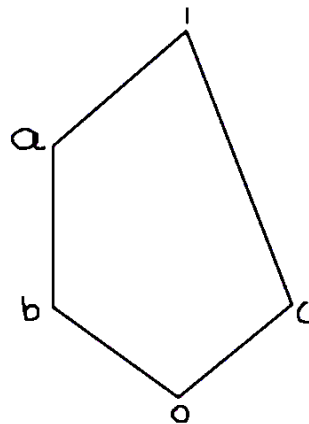


- Let $A = \{1, 2, 3, 4\}$, $B = \{w, x, y, z\}$ and $C = \{p, q, r, s\}$. Consider $R_1 = \{(1, x), (2, w), (3, z)\}$ a relation from A to B . $R_2 = \{(w, p), (z, q), (y, s), (x, p)\}$ a relation from B to C . (i) compute $R_1 \circ R_2$ from A to C . (ii) Write $M(R_1)$, $M(R_2)$, $M(R_1 \circ R_2)$ (iii) Verify $M(R_1) \cdot M(R_2) = M(R_1 \circ R_2)$ or not? Also illustrate (iii) graphically. CO2 (07)

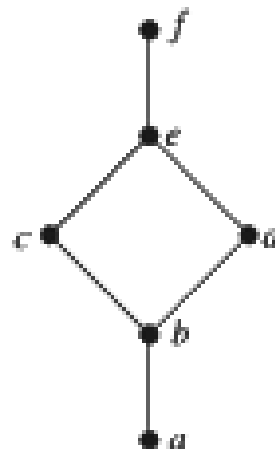
- c) Determine whether the posets with Hasse diagram are lattice or not. CO2 (07)
Justify.



4. a) Determine whether the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = |x| + x, \forall x \in \mathbb{R}$ is one to one and onto or not. CO2 (06)
- b) Define a partially order on a set A. Given a partially order $\{(a,b)/a \text{ divides } b\}$ on $A = \{1,2,3,4,6,8,12\}$. Draw the digraph of the partially ordering relation on A. Also draw the Hasse diagram for the relation. CO2 (07)
- c) Verify whether the following lattice's are Boolean algebra or not? In either of the case explain. CO2 (07)



i)



ii)

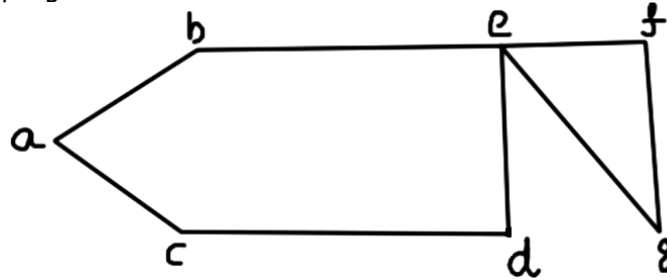
UNIT - III

5. a) State: (i) Sum and product rule. (ii) Principle of inclusion-exclusion (iii) Pigeon hole principle. CO3 (06)
- b) Define recurrence relation. Solve $a_n - 6a_{n-1} + 9a_{n-2} = 0, n \geq 2$ given $a_0 = 5, a_1 = 12$. CO3 (07)
- c) A man has 11 close relatives and he wishes to invite 5 of them for dinner. In how many ways can he invite them in the following manner. (i) there is no restriction on the choice. (ii) Two particular persons will not attend separately. (iii) Two particular persons will not attend together. CO3 (07)

6. a) Show that if seven numbers from 1 to 12 are chosen then two of them will be add up to 13. CO3 (06)
- b) Solve the Fibonacci relation $F_{n+2} = F_{n+1} + F_n$, $n \geq 0$. Given that $F_0 = F_1 = 1$. CO3 (07)
- c) In how many ways can 4a's, 3b's and 2c's be arranged so that all the identical letters are not in the same block? CO3 (07)

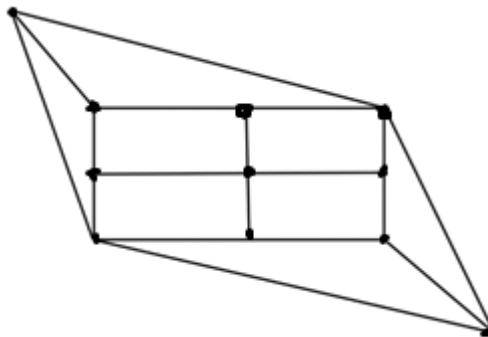
UNIT- IV

7. a) In the graph given below: CO4 (06)

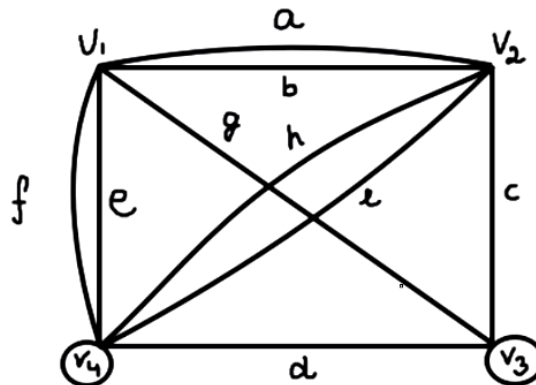


Determine (i) Walk from b to b that is not a trail. (ii) b-d trail that is not a path. (iii) a path from b to d. (iv) a closed walk from b to b that is not a circuit. (v) a circuit from b to b that is not a cycle. (vi) a cycle from b to b.

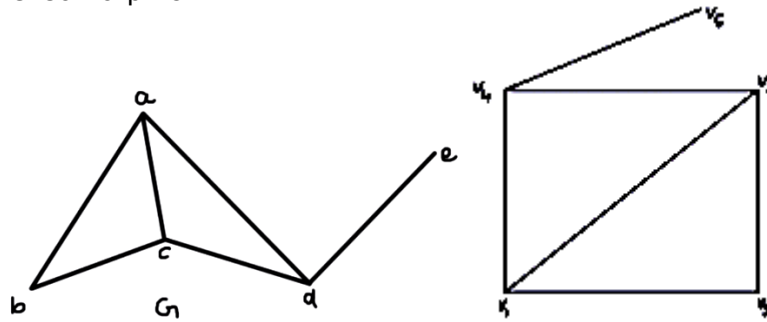
- b) (i) Define complement of a graph and give an example. CO4 (07)
- (ii) Determine $|V|$ for the following graphs:
- G has 9 edges and all vertices has degree 3.
 - G has 10 edges with 2 vertices of degree 4 and all other of degree 3.
- c) (i) Define colouring and chromatic number of a graph. CO4 (07)
- (ii) Find chromatic number of the following graph with the help of a coloured graph.



8. a) Define adjacency and incidence matrix of a graph and write the two matrices for the following graph. CO4 (06)



- b) Define isomorphism of the two graphs. Verify whether the graphs given below are isomorphic. CO4 (07)



- c) Define Euler's graph and Hamiltonian graph with an example for each. CO4 (07)
Also give an example of a graph which is Euler but not a Hamiltonian graph.

UNIT - V

9. a) i) Define Monoid with an example. CO5 (06)
ii) Prove that in a group, identity element is unique.
Prove that the set of real numbers other than 1 is an abelian group with CO5 (07)
b) respect to the binary composition '*' defined by $a * b = a + b - ab$.
c) State and prove Fermat's theorem. CO5 (07)
10. a) Let G be a group of integers under the operation addition. Let CO5 (06)
 $H = \{3x / x \in \mathbb{Z}\}$. Is H a subgroup of G .
b) Let $(\mathbb{Z}, +)$ be the additive group of integers. Show that the function CO5 (07)
 $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(a, b) = a + b, \forall a, b \in \mathbb{Z}$ is a isomorphism.
c) Using Fermat's theorem, solve $27x \equiv 11 \pmod{7}$. CO5 (07)
